

Machine Learning Based Performance Prediction of Analog CMOS Circuits

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partial fulfillment of the requirements for the award of
the degree of*

Bachelor of Technology In Electronics and Communication Engineering

by

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Abstract

Analog CMOS circuit design plays a vital role in modern electronic systems including communication devices, RF systems, biomedical electronics, embedded systems, and mixed-signal integrated circuits. Conventional analog circuit design requires extensive parameter tuning, repeated simulations, and manual optimization to achieve desired specifications such as voltage gain, bandwidth, noise figure, and power consumption. As the complexity of integrated circuits increases, traditional approaches become computationally expensive and highly time-consuming.

This project proposes a Machine Learning-based Analog CMOS Circuit Performance Prediction system that automates analog design parameter estimation using data-driven techniques. Three important analog CMOS circuits were selected for the study:

- Common Source Voltage Amplifier (CSVA)
- Cascode Voltage Amplifier (CVA)
- Gilbert Cell Mixer

The circuits were designed and simulated in Cadence Virtuoso. Various circuit parameters such as transistor widths, supply voltage, gate bias voltage, load resistance, and current values were swept over predefined ranges to generate datasets. The generated datasets were then preprocessed and used to train machine learning models for predicting transistor sizing parameters and performance metrics.

Among different machine learning algorithms, the Random Forest Regressor was selected because of its high prediction accuracy, robustness against overfitting, and ability to model nonlinear analog circuit behavior.

The trained model was capable of predicting parameters such as transistor W/L ratio, gain, bandwidth, and power consumption with acceptable accuracy.

A web-based application was also developed where users can upload circuit datasets and automatically obtain predictions using trained models. The web platform improves usability and simplifies analog design exploration.

The project demonstrates that machine learning can significantly reduce manual parameter sweeping, accelerate analog circuit design workflows, and provide efficient performance estimation for CMOS analog circuits.

Introduction

Overview of Analog CMOS Circuits

Analog CMOS circuits are fundamental building blocks in modern electronics. Unlike digital circuits that operate using discrete logic levels, analog circuits process continuous-time signals and are widely used in amplification, filtering, modulation, oscillation, and signal conversion applications.

Analog CMOS circuits are extensively used in:

- Wireless communication systems
- RF front-end circuits
- Audio amplifiers
- Sensor interfaces
- Biomedical devices
- Mixed-signal systems
- Phase-locked loops
- Signal processing circuits

The increasing demand for high-speed and low-power integrated circuits has made analog circuit optimization extremely important.

Need for Analog Circuit Optimization

Traditional analog circuit design involves selecting transistor dimensions and biasing conditions manually through repeated simulations. Designers usually perform parameter sweeping to identify suitable operating points that satisfy performance requirements such as:

- Voltage gain
- Bandwidth
- Noise figure
- Power dissipation

- Conversion gain
- Linearity

As circuit complexity increases, the design space becomes extremely large. Exploring every possible combination manually requires significant computational effort and expert knowledge.

For example, even a simple analog amplifier may contain multiple design variables such as:

- NMOS width
- PMOS width
- Load resistance
- Bias current
- Supply voltage
- Channel length

Each parameter affects multiple performance metrics simultaneously, making optimization difficult.

Role of Machine Learning in Analog IC Design

Machine Learning provides an efficient solution to analog design challenges by learning the relationship between circuit specifications and circuit parameters directly from simulation data.

Instead of manually sweeping parameters repeatedly, ML models can predict transistor dimensions and performance metrics automatically based on previously trained datasets.

Advantages of ML-assisted analog design include:

- Reduced simulation time
- Faster design exploration
- Automated parameter prediction

- Improved productivity
- Reduced manual effort
- Intelligent optimization

Recent research such as the AI Circuit benchmark dataset demonstrated the effectiveness of ML techniques in analog and RF circuit parameter prediction.

Objectives

The primary objective of this project is to develop a machine learning-based system for predicting analog CMOS circuit parameters and performance metrics.

The detailed objectives are:

1. To design analog CMOS circuits using Cadence Virtuoso.
2. To simulate **Common Source Voltage Amplifier, Cascode Voltage Amplifier, and Gilbert Cell Mixer** circuits.
3. To generate datasets by sweeping important design parameters.
4. To preprocess simulation data for machine learning training.
5. To train machine learning models for analog parameter prediction.
6. To predict transistor sizing parameters such as Width/Length ratio.
7. To analyze analog performance metrics including:
 - Voltage gain
 - Bandwidth
 - Power consumption
 - Conversion gain
8. To develop a user-friendly web application for prediction automation.
9. To reduce manual parameter sweeping in analog IC design workflow.

Literature Review

Traditional Analog Design Techniques

Conventional analog IC design relies heavily on expert knowledge and repeated simulations. Designers manually tune transistor parameters until the required specifications are achieved.

The traditional design flow includes:

1. Circuit topology selection
2. Initial parameter estimation
3. Schematic design
4. Parameter sweeping
5. Simulation verification
6. Performance optimization

Although effective, this process becomes highly inefficient for complex circuits.

Early Machine Learning Approaches

Researchers began exploring machine learning methods for circuit optimization to reduce design complexity.

Initial approaches included:

- Artificial Neural Networks
- Genetic Algorithms
- Evolutionary Optimization
- Bayesian Optimization

These techniques attempted to estimate analog performance metrics using previously simulated datasets.

However, early methods suffered from:

- Limited dataset size
 - Poor generalization
 - High computational complexity
 - Inability to model nonlinear behavior accurately
-

Modern ML-Based Analog Design

Recent developments in machine learning introduced advanced methods such as:

- Random Forest Regression
- Deep Neural Networks
- Reinforcement Learning
- Graph Neural Networks
- Transformers

The AICircuit benchmark dataset introduced a comprehensive framework for AI-driven analog IC design containing multiple homogeneous and heterogeneous analog circuits.

The research demonstrated that machine learning models can learn mappings between:

- Performance metrics
- Transistor parameters
- Circuit configurations

Modern ML-assisted design directly predicts circuit parameters from desired specifications:

$$x = M(y)$$

Where:

- x = circuit parameter vector
- y = performance metric vector

- M= machine learning model

This eliminates repeated parameter sweeping.

Methodology

Overall Workflow

The project workflow consists of the following stages:

1. Analog Circuit Design
2. Parameter Selection
3. Parameter Sweeping
4. Cadence Simulation
5. Dataset Generation
6. Data Preprocessing
7. Machine Learning Model Training
8. Prediction and Evaluation
9. Web Application Development

Step 1: Circuit Design

Three analog circuits were designed in Cadence Virtuoso:

Common Source Voltage Amplifier (CSVA)

A single-stage analog amplifier used for voltage amplification.

Cascode Voltage Amplifier (CVA)

A two-transistor amplifier configuration providing high gain and bandwidth.

Gilbert Cell Mixer

A differential RF circuit used for frequency conversion.

Step 2: Parameter Selection

Important analog design parameters were selected:

- Transistor width (W)
- Channel length (L)
- Supply voltage (VDD)
- Gate voltage (VGS)
- Load resistance (RD)

- Bias current

These parameters strongly influence analog performance metrics.

Step 3: Parameter Sweeping

Each parameter was swept across predefined ranges.

Example:

- $V_{DD} = 1.2V$ to $2V$
- $W_n = 3\mu m$ to $10\mu m$
- $R_D = 600\Omega$ to 2500Ω

This produced multiple simulation combinations.

Step 4: Cadence Simulation

Cadence Virtuoso was used for:

- DC analysis
- AC analysis
- Transient analysis

Simulation outputs included:

- Gain
 - Bandwidth
 - Power consumption
 - Conversion gain
-

Step 5: Dataset Generation

Simulation outputs were stored in CSV format.

Each dataset row contained:

- Input parameters
- Output performance metrics

The dataset was divided into:

- Training data
- Testing data

Circuit Design

1.COMMON SOURCE VOLTAGE AMPLIFIER

The Common Source Voltage Amplifier is one of the most widely used analog amplifier configurations.

In this circuit:

- Input is applied to the gate terminal
- Output is obtained at the drain terminal
- Source terminal is grounded

The CSVA provides:

- High voltage gain
- Simple structure
- Moderate bandwidth

Input Parameters

- Supply voltage
- Gate voltage
- NMOS width
- Load resistance

Output Metrics

- Voltage gain
- Bandwidth
- Power consumption

Applications:

- Audio amplification
- Analog signal processing
- RF front-end stages

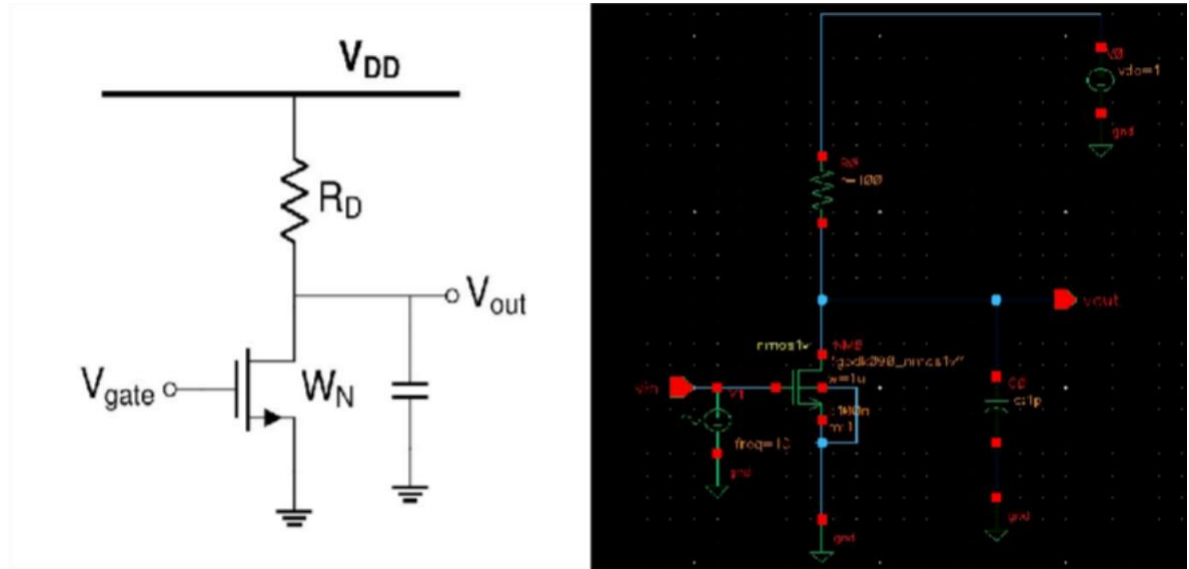


Fig.1 Schematic of CSVA (Circuit diagram and virtuoso schematic)

2. CASCODE VOLTAGE AMPLIFIER

The Cascode Voltage Amplifier combines:

- Common source stage
- Common gate stage

Advantages:

- High gain
- Improved bandwidth
- Reduced Miller effect
- High output resistance

The cascode configuration is commonly used in:

- RF circuits
- High-frequency amplifiers
- Analog front-end systems

Performance Metrics

- Voltage gain
- Noise figure
- Power consumption

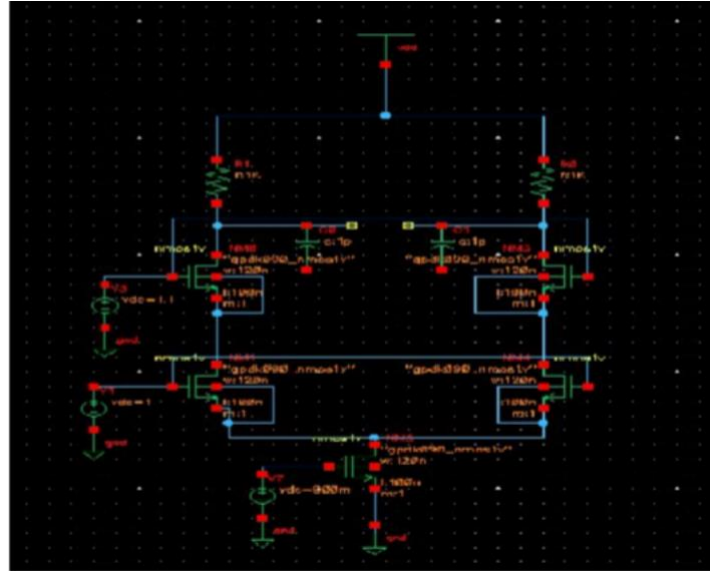
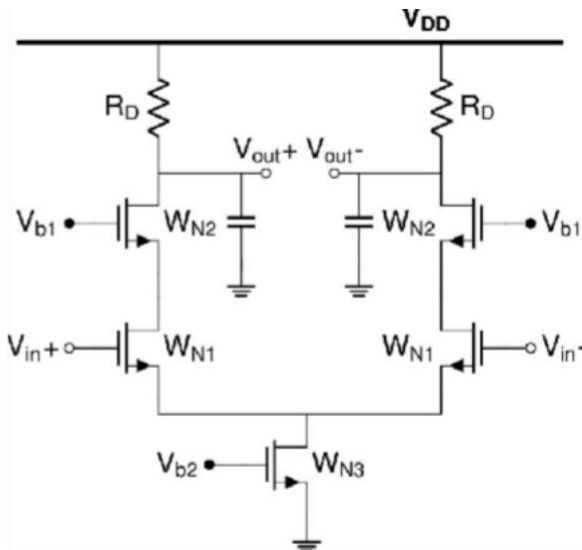


Fig 2. Schematic of CVA (Circuit diagram and virtuoso schematic)

3. GILBERT CELL MIXER

The Gilbert Cell Mixer is a differential analog multiplier used in RF communication systems.

Functions:

- Frequency conversion
- Signal modulation
- Signal demodulation

Advantages:

- High linearity
- Differential operation
- Low distortion

Applications:

- Wireless communication
- RF receivers
- Transmitters

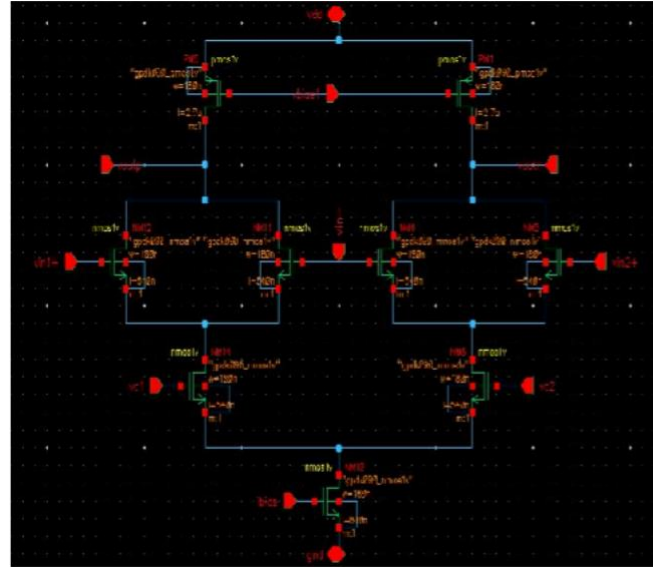
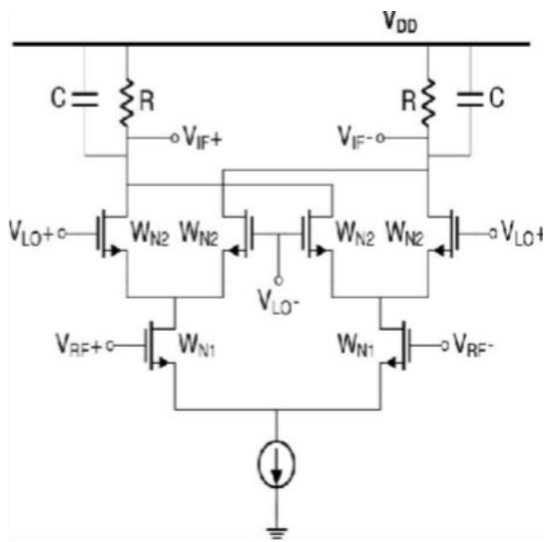


Fig3. Gilbert Cell Schematic (Circuit diagram and virtuoso schematic)

Dataset Generation

Datasets were generated by sweeping circuit parameters across predefined ranges in Cadence Virtuoso. Each simulation result was stored as a row containing:

- Circuit Parameters
- Transistor Dimensions
- Performance Metrics

The datasets were divided into:

- Training Dataset
- Testing Dataset

The approach follows the dataset generation methodology proposed in AICircuit.

Machine Learning Model

Random Forest Regressor

Random Forest is an ensemble machine learning algorithm consisting of multiple decision trees.

Advantages:

- High prediction accuracy
- Handles nonlinear relationships
- Robust against overfitting
- Fast training

The model predicts:

- Transistor W/L ratio
- Gain
- Bandwidth
- Power consumption

Training Process

Steps:

1. Load dataset
2. Normalize data

3. Split train-test data
4. Train Random Forest model
5. Evaluate prediction accuracy

Libraries used:

- Scikit-learn
- Pandas
- NumPy

Website Development

A web-based platform was developed for user interaction.

Features

Dataset Upload

Users can upload CSV files.

Automatic Model Selection

The system identifies circuit type automatically.

Prediction Output

The website predicts:

- W/L ratio
- Gain
- Bandwidth
- Power consumption

Interactive UI

The interface was designed with modern UI/UX principles.

Technologies Used

Backend:

- Python
- Flask

Frontend:

- HTML

- CSS
- JavaScript

ML Libraries:

- Scikit-learn
- Pandas
- NumPy

Results and Analysis

Sample CSVA Prediction Results

VDD	Vgate	Wn	Rd	Bandwidth	Gain	Power
1.2	0.60	3	2500	90	11.8	0.213
1.4	0.75	5	1800	120	13.2	0.547
1.5	0.80	6	1500	180	14.5	0.741
1.8	0.90	8	900	280	17.1	1.731

Analysis

The Random Forest model successfully learned the nonlinear relationships between:

- Circuit parameters
- Performance metrics

Observations:

- Gain increases with transistor width
- Power increases with supply voltage
- Bandwidth varies with load resistance

The prediction accuracy was acceptable for analog design estimation.

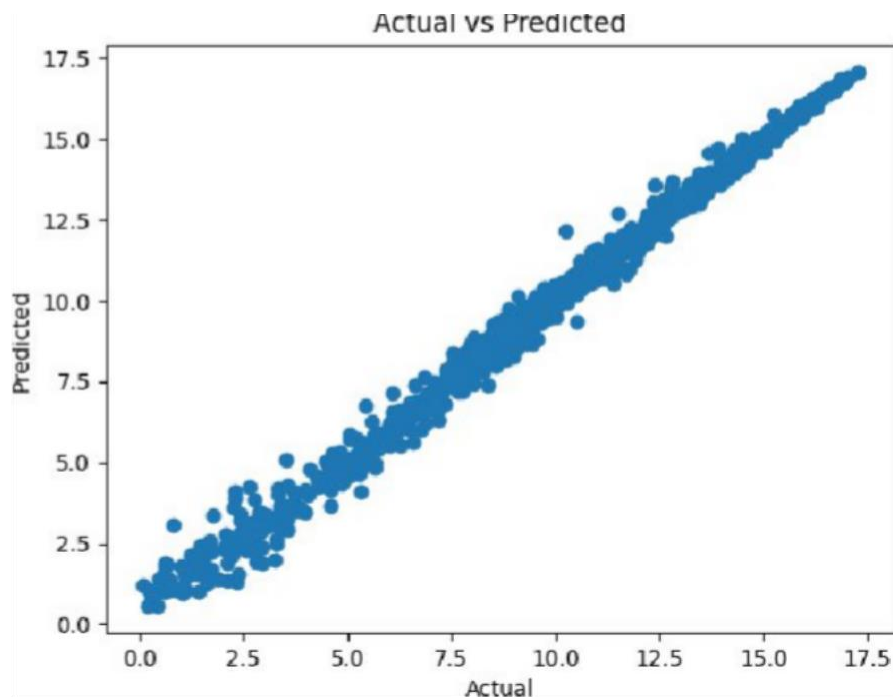


Fig.1 Actual vs Predicted graph of CSVA Circuit with Random forest Alogrithm

Advantages

1. Faster analog circuit optimization
2. Reduced manual tuning
3. Lower simulation time
4. Efficient parameter prediction
5. Supports automated design exploration
6. Easy integration with EDA tools

Challenges

High Dimensionality

Large parameter spaces increase training complexity.

Nonlinear Analog Behavior

Analog circuits exhibit highly nonlinear characteristics.

Data Scarcity

Generating simulation datasets is time-consuming.

Generalization

Models may not generalize well to unseen topologies.

Applications

- Analog IC Design
- RF Circuit Optimization
- AI-assisted EDA
- Low-power CMOS design
- Communication systems
- Mixed-signal ICs

Future Scope

- Integration with deep learning models
- Real-time Cadence integration
- Cloud-based EDA systems
- Support for larger analog systems
- Reinforcement learning-based optimization

Conclusion

The project successfully demonstrated the use of machine learning for analog CMOS circuit parameter prediction.

Three analog circuits:

- CSVA
- CVA
- Gilbert Cell Mixer

were designed and simulated using Cadence Virtuoso.

Datasets generated from simulations were used to train a Random Forest Regressor capable of predicting:

- W/L ratio
- Gain
- Bandwidth

- Power consumption

A web-based platform was also developed for automated prediction.

The results prove that machine learning can significantly reduce manual parameter sweeping and improve analog IC design efficiency.

References

1. AICircuit: A Multi-Level Dataset and Benchmark for AI-Driven Analog Integrated Circuit Design, 2024.
2. Cadence Virtuoso Spectre Circuit Simulator Reference, 2023.
3. Behzad Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill, 2016.
4. Leo Breiman, "Random Forests," Machine Learning Journal, 2001.
5. G. Gielen and R. Rutenbar, "Computer-Aided Design of Analog and Mixed-Signal Integrated Circuits," Proceedings of IEEE, 2000.